

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the —README.md— for this assignment includes instructions to regenerate this handout with your typeset $\text{\LaTeX}$ solutions.

2.ci

Mean: 0.568, Std: 0.063

2.cii

Table 1: LinUCB with "Popular" Arm Updates

K Value	Mean	Std
500	0.441	0.082
1000	0.490	0.080
2500	0.535	0.073
5000	0.556	0.066

The "Popular" strategy seems worse than the baseline. I think it's because this strategy just reinforce the existing model trend without considering the actual user preference.

2.ciii

Table 2: LinUCB with "Corrective" Arm Updates

K Value	Mean	Std
500	0.441	0.082
1000	0.490	0.080
2500	0.535	0.073
5000	0.556	0.066

It's interesting that "Corrective" has identical results as "Popular". My guess is that the most popular arms may also be the most incorrectly predicted arms. This will results in the same effect as Popular strategy, where the model will reinforce the existing trend and thus converge to the same result.

2.civ

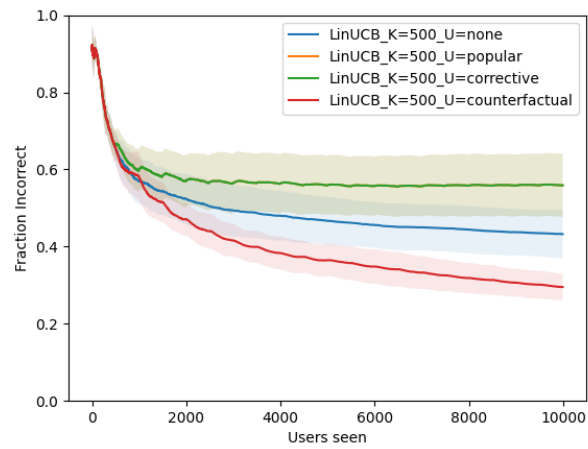
Table 3: Comparison of Mean Total Fraction Correct Across Strategies

K Value	None	Popular	Corrective	Counterfactual
500	0.568	0.441	0.441	0.705
1000	0.568	0.490	0.490	0.670
2500	0.568	0.535	0.535	0.618
5000	0.568	0.556	0.556	0.570

The counterfactual strategy is clearly the best of the three arm update strategies. It is the only strategy that consistently performs better than the baseline case where no new arms are added.

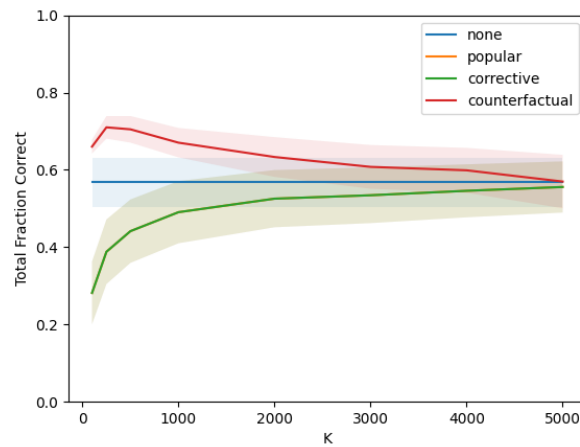
I think it's because counterfactual is the only strategy that considers all user's preferences.

2.cv



The counterfactual strategy shows the fastest learning curve and the corrective and popular strategies perform worse than the none baseline.

2.cvi



Getting worse: The counterfactual perform worse when K is getting large. I think this is because smaller values of K allow for more frequent adjustments based on recent user interactions. **Getting better:** The popular and corrective strategies improve as K increases. I think it's because larger K provides a more stable and less noisy signal of what is truly "popular" or where "corrective" action is most needed.

2.d

Ethical issue: For scenario (1), whether this has ethical problem is highly depending on what the user preferences are. In scenario (1), eventually, there will be more and more creators that only creates contents that meets most of the user preferences. The diversity of the content will be very low. If this major trend is harmful, then it's a big problem.

Scenario (2) has a similar issue but worse. It's because the algorithm will shift the user preferences to the popular ones and making niche contents has less and less opportunities to be seen.

Scenario (3) is even worse. Instead of randomly selecting contents as scenario (2) doing, the popular contents might be manipulated.

Scenario (4) kind of mitigate the problem where users can intervening the algorithm.

Distinguishing between these scenarios would require specific experiments beyond analyzing standard logs.

2.e

I think the platform is primarily responsible for identifying and mitigating negative feedback loops because they have direct control to the feed algorithms and they has more context to identify possible issues. However, all the roles in this loop should be able to recognize and have some ways to report the issues to the platform.